

Module 1: Keplerian Orbit Mechanics and Satellite Subsystems

A communication satellite consists of a space segment and ground segment.

Keplerian orbital parameters define the satellite trajectory including semi-major axis, eccentricity, inclination, argument of perigee, and right ascension of ascending node.

Geostationary earth orbits operate at altitude of 35786 kilometers with orbital period of 24 hours.

The satellite payload comprises transponders, multiplexers, and high-gain parabolic antennas.

Spacecraft telemetry tracking and command subsystems maintain proper satellite attitude and orbital s